

DOK 3 TASK QUESTIONS

Reverse-engineer a polynomial from its graph

DOK arc for the day: gradual-release Launch (I-do $\rightarrow$ you-do $\rightarrow$ I-check $\rightarrow$ we-do-together) gives students a worked example to lean on, THEN the DOK-3 Mystery Graph hits in Explore, then DOK-2 consolidation, then a DOK-1 cool-down at Exit. The DOK-3 task is loaded near the front of Explore (not buried at the end) so the reasoning happens while energy is high.

- **Entry / Access (build understanding, low floor)**

- Where does this graph cross the x -axis? Where does it touch without crossing?
- What is the lowest possible degree this graph could be? Why?
- What does the end behavior tell you about the leading coefficient?
- If a graph touches but does not cross at $x = 2$, what kind of factor must $(x - 2)$ be?

- **Developing (connect representations, reasoning)**

- How do the zeros and their multiplicities together determine the shape of the graph near each zero?
- Why does an even multiplicity produce a touch and an odd multiplicity produce a cross?
- Two students propose different factored forms for the same graph. How would you decide which (if either) is correct?

- **Core Task (DOK 3 – reasoning & justification, FRONT-LOADED)**

- Given only a graph (zeros at $x = -1$ touch, $x = 3$ cross; ends down-down), construct a possible factored form. Justify each factor.
- Could there be more than one polynomial with this graph? What is the same about all of them, and what can vary?
- Defend your leading coefficient sign using end behavior, not by plugging in numbers.
- A classmate writes $f(x) = (x + 1)(x - 3)$ for the same graph. Build a counterexample showing why their answer fails the touch-at- -1 requirement.

- **Extension (higher-level thinking, generalization)**

- Generalize: from any graph with labeled touches, crosses, and end behavior, can you always recover a factored form (up to a constant)?
- How would the answer change if the touch at $x = -1$ were a multiplicity-4 touch instead of multiplicity-2? What would the graph look like differently?

- **Discussion / Argumentation prompts (to support collaboration)**

- Do you agree with your partner's factored form? Where exactly do you disagree?
- What evidence from the graph best supports your factor for $x = -1$?
- Can there be more than one reasonable approach? Compare them side-by-side.

Rationale: Students learn to read multiplicity off a graph through gradual release (I-do, you-do, I-check, we-do-together), then apply the rule by reverse-engineering a polynomial from a graph in a DOK-3 task at the start of Explore. The worked-example scaffold de-risks the reasoning task while preserving the productive-struggle moment.

Materials: L35_P3_obs student packet, projected “Mystery Graph” slide, sentence-frame card, Desmos available for the share-out.

Date: 5/8/26

Grade Level: 11

Subject: Algebra 2 – Lesson 3-5, Day 3 / Period 3 Observation (Multiplicity + Factored-Form Graphing)

Common Core State Standards:

- **HSA.APR.B.3:** Identify zeros of polynomials when suitable factorizations are available, and use the zeros to construct a rough graph of the function defined by the polynomial.
- **HSF.IF.C.7.C:** Graph polynomial functions, identifying zeros when suitable factorizations are available, and showing end behavior.

Knowledge Needed:

- Factored form of a polynomial $\rightarrow$ zeros connection
- End behavior depends on degree parity and leading-coefficient sign
- Tier 2 vocabulary: zero, root, multiplicity, factor, end behavior, degree, leading coefficient

Skills Needed:

- Read x -intercepts from a graph
- Distinguish a touch from a cross visually
- Convert factored form to expanded form (and back) for verification

Standards for Mathematical Practice

1. Make sense of problems and persevere in solving them
3. Construct viable arguments and critique the reasoning of others
7. Look for and make use of structure

Unit Goals: Multiplicity is the bridge between symbolic factored form and graph behavior. Students who own this connection can later read rational functions, exponential models, and trig graphs structurally rather than by memorization.

Content Objective: 1. Construct a factored polynomial from a graph by reasoning about zeros, multiplicity, and end behavior. 2. Justify why even multiplicity produces a touch and odd multiplicity produces a cross at a zero.	Criteria for Success/Formative Assessment: • TPS check-ins during Launch and Explore (4–6 circulation laps) • Exit ticket: vocabulary + 1 quick reverse-engineer Self-Reflection Assessment: • “The biggest thing I learned today was ____.” Teacher Assessment • Exit-ticket review before P3 (Storage Box) on Mon 5/11	
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Language Objective: Students will justify a chosen factored form from a graph, verbally and in writing, using sentence frames and structured think-pair-share.	
Modifications or Support for IEP students. - Rephrasing/CFU of expectations - Pre-printed graph copy taped into packet for annotation - Extended TPS think-time on cue	Supports for ELL students - Vocabulary card: zero, root, multiplicity, touch, cross, end behavior - Sentence starters during TPS: “The factor for $x = \underline{\hspace{1cm}}$ must be $\underline{\hspace{1cm}}$ because the graph $\underline{\hspace{1cm}}$” “The leading coefficient is $\underline{\hspace{1cm}}$ because the ends go $\underline{\hspace{1cm}}$” “I notice $\underline{\hspace{1cm}}$, so I think $\underline{\hspace{1cm}}$” - Two-reads strategy on the Mystery Graph prompt

Lesson Sequence

Lesson Part	Teachers will...	Students will...	Questions to consider...
Do Now (5 min) Quick warm-up, NOT the DOK-3 hook	Project 3 factored polynomials. For each, list the zeros (no graphing yet). $f(x) = (x - 2)(x + 5)$ [zeros: 2, -5] $g(x) = (x - 1)^2(x + 3)$ [zeros: 1, -3] $h(x) = x(x - 4)(x + 2)$ [zeros: 0, 4, -2] Adult role: Spot-check; flag #2 (the squared factor) for the Launch hook.	Read each polynomial; list zeros; circle anything unusual.	How can you engage students in the lesson? How are all adults intentionally planned for?

Lesson Part	Teachers will...	Students will...	Questions to consider...
Launch (15 min): <i>Gradual release</i> • I do • You do • I check • We do together	I do (4 min). Project $f(x) = (x - 2)^2(x + 3)$. Think aloud while sketching: “Zero at $x = 2$, factor squared $\rightarrow$ even multiplicity $\rightarrow$ touches. Zero at $x = -3$, factor to the first power $\rightarrow$ odd multiplicity $\rightarrow$ crosses. Degree 3, leading coefficient +, so end behavior down-up.” Name the touch/cross rule explicitly on the board. You do (4 min). Project $g(x) = (x + 1)(x - 4)^2$. Students sketch independently in their packet using the rule. I check (3 min). Circulate. Pull two student sketches to project (one correct, one with a common error like swapping touch/cross). Name the misconception out loud, fix it as a class. We do together (4 min). Project $h(x) = -x(x + 2)^2(x - 1)$ (one harder: leading coefficient negative, three zeros). Whole-class call-and-response: “Where are the zeros?” “Which one touches?” “Why does it open downward?” Teacher writes; students copy. Read content + language objectives aloud at the end. Adult role: Model precisely in I-do; circulate aggressively in You-do; name errors without judgment in I-check; conduct We-do as a chorus.	I do: Watch and annotate the example in the packet; copy the touch/cross rule. You do: Independently sketch $g(x) = (x + 1)(x - 4)^2$ using the rule. I check: Compare with the projected exemplar; revise; participate in the misconception fix. We do together: Call out answers chorally for the harder example; copy the final sketch.	What will you explicitly model and how? How will you know if students are ready to work alone? How are all adults intentionally planned for?

Lesson Part	Teachers will...	Students will...	Questions to consider...
Explore (25 min): DOK-3 task, then DOK-2 consolidation • Mystery Graph (DOK-3, 10 min) • Mixed graph $\leftrightarrow$ factored practice (DOK-2, 15 min) • Students doing the heavy lifting	Mystery Graph (10 min, TPS). Project: zeros at $x = -1$ (touch), $x = 3$ (cross); ends both pointing <i>down</i>. Pose: “Build a possible factored form for this graph. Justify every piece – the factors, their powers, and the leading coefficient.” Note for students: this is harder than Launch – the ends-down case forces you to reason about leading coefficient AND degree together. Start TPS – 3 min independent, 3 min partner share, 4 min whole-group. Circulate; tag 2–3 different responses. In whole-group share, press: “Why must the leading coefficient be negative?” “Could the degree be 4 instead of 3? What would change?” DOK-2 consolidation (15 min, TPS). Hand out 3 practice items from L35 P2 bank: – Item 1: factored $\rightarrow$ graph – Item 2: factored $\rightarrow$ graph (with a multiplicity-3 factor) – Item 3: graph $\rightarrow$ factored (lower-stakes than Mystery Graph) Questions while circulating: • Where does the graph touch vs. cross? Why? • What does end behavior tell you about degree and sign? • How would you check your factored form is right? Adult role: 4–6 circulation laps; pull a struggling pair to a re-teach huddle if needed; tag exemplars for Share.	Mystery Graph: TPS – propose a factored form; defend each factor and the leading coefficient using the sentence frame. DOK-2 items: work in pairs; record reasoning, not just answers; verify with Desmos at the end of each item.	What will students do for most of the session? What will they practice? How? What type of academic conversations? How will you be interacting with them?

Lesson Part	Teachers will...	Students will...	Questions to consider...
Share/Summary (5 min) • Synthesis • Return to objectives • Name the rule again	Re-state the multiplicity rule using student language from Launch. Press: “Where in today’s items did the rule save you time?” Project objectives. Ask: “Which one did you hit? Where did you struggle?”	Share one item where the rule helped; agree/disagree/add on; reflect on objectives.	How will students express what they have learned?
Exit Ticket (5 min) <i>DOK-1 cool-down</i>	Project exit prompt (low-stakes by design): 1. A factor of $(x - 4)^2$ produces a ____ at $x = 4$ (touch / cross). 2. A factor of $(x + 1)$ produces a ____ at $x = -1$ (touch / cross). 3. One sentence: “The biggest thing I learned today was _____.”	Complete exit ticket independently; turn in on the way out.	How will students show what they’ve learned based on the objective?